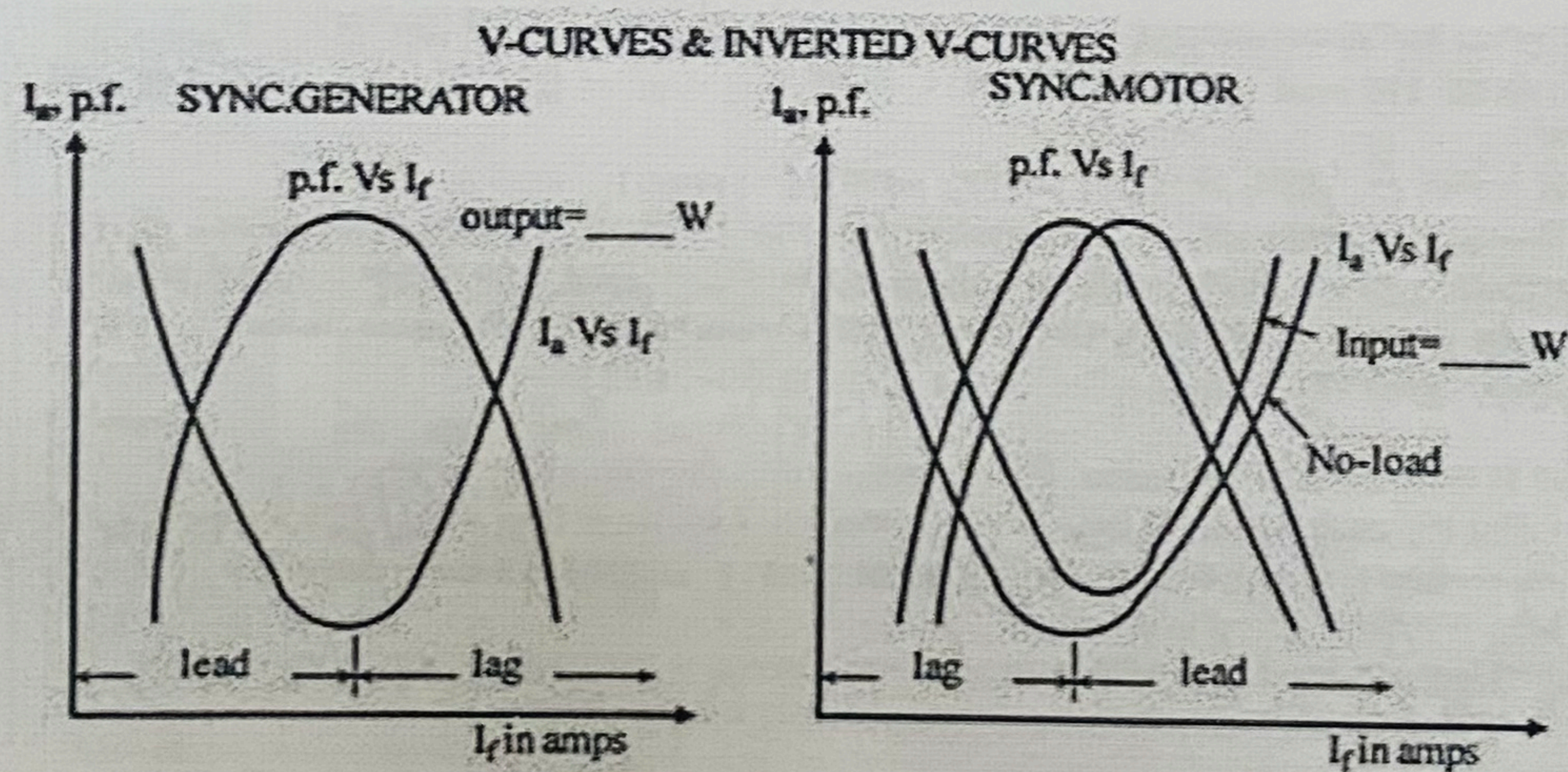




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Note: a) Over-excited synchronous generator operates at lagging p.f. and under-excited synchronous generator at leading pf and b) over-excited synchronous motor operates at leading pf and under-excited synchronous motor at lagging pf

MODEL GRAPH



SAMPLE CALCULATION (Set No. ____)

Generator delivering an output of ____ W

$V = 400\text{V}$, $I_a = \text{____ A}$, $I_f = \text{____ A}$, $I_{amin} = \text{____}$

$\cos \phi = \frac{I_{amin}}{I_a} = \text{____}$ lag (/lead) (Note : under-excited $\rightarrow$ lead & over-excited $\rightarrow$ lag)